

EMERSON, MB, Canada

WMO: 715600

Lat: 49.0014N

Lon: 97.2417W

Elev: 242

StdP: 98.45

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.4	-28.0	-34.6	0.1	-30.1	-32.2	0.2	-27.9	16.1	-14.2	14.3	-14.7	3.2	320	0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	30.4	21.4	28.8	20.6	27.3	19.6	23.6	28.3	22.2	26.9	21.0	25.6	5.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	17.2	26.8	20.6	15.7	25.2	19.3	14.4	23.7	72.0	28.4	66.6	27.0	62.0	25.6	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.3	10.0	-33.8	33.7	3.2	0.9	-36.1	34.4	-38.0	34.9	-39.8	35.4	-42.1	36.1		
(5)			-33.9	25.9	3.2	1.2	-36.2	26.8	-38.1	27.5	-39.9	28.2	-42.2	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-14.9	-13.0	-5.1	4.8	12.2	17.5	20.1	18.8	14.1	6.0	-2.8	-11.6
	DBStd	13.74	7.95	7.64	7.64	6.00	5.05	3.53	3.09	3.38	4.64	5.08	6.32	7.66
	HDD10.0	3312	771	645	468	173	36	1	0	0	14	147	386	671
	HDD18.3	5437	1029	878	726	406	199	56	18	36	140	384	635	929
	CDD10.0	1088	0	0	1	17	103	225	312	272	138	21	1	0
	CDD18.3	172	0	0	0	0	7	31	71	49	14	1	0	0
	CDH23.3	1773	0	0	0	6	125	301	653	512	164	12	0	0
	CDH26.7	456	0	0	0	0	35	68	171	141	39	2	0	0
(14) Wind	WSAvg	4.7	5.1	4.9	5.0	5.2	5.1	4.3	3.5	3.6	4.3	4.9	4.9	4.9
(15) Precipitation	PrecAvg	488	12	10	19	29	62	91	76	65	53	36	18	14
	PrecMax	756	40	37	75	84	168	229	216	206	156	126	99	33
	PrecMin	273	0	0	0	0	12	14	12	12	2	1	0	1
	PrecStd	108	10	9	17	22	37	51	45	45	35	29	20	8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.2	5.3	16.5	23.8	30.7	30.8	32.1	32.7	30.2	24.9	14.6	4.9
		MCWB	1.6	3.8	11.8	12.4	17.2	20.9	23.5	22.8	19.6	15.6	9.7	2.3
	2%	DB	1.0	2.6	10.5	20.5	26.6	28.6	30.2	30.0	27.5	20.5	10.8	2.0
		MCWB	0.2	1.5	7.4	11.5	15.3	19.6	22.5	21.6	18.9	13.8	7.3	0.7
	5%	DB	-1.4	0.7	6.8	17.6	23.9	26.7	28.6	28.0	25.1	17.1	8.2	0.3
		MCWB	-2.2	0.0	4.7	10.0	14.2	18.8	21.6	20.4	17.3	11.9	5.3	-0.6
	10%	DB	-3.6	-1.5	3.8	14.8	21.4	24.8	27.0	26.2	22.5	14.2	5.3	-1.4
		MCWB	-4.2	-2.2	2.4	8.7	13.5	17.8	20.6	19.2	16.2	10.0	3.4	-2.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	4.3	12.3	14.5	19.6	23.9	25.5	25.0	22.3	17.3	10.8	2.5
		MCDB	2.7	5.4	14.9	20.8	25.6	28.4	30.1	29.8	27.9	22.5	13.7	4.5
	2%	WB	0.1	1.6	7.8	12.2	17.5	21.6	23.9	22.8	20.2	14.6	7.7	0.9
		MCDB	0.8	2.4	10.6	18.5	22.8	26.1	28.5	28.3	25.0	19.7	10.3	1.9
	5%	WB	-2.2	0.0	4.8	10.6	16.0	20.1	22.8	21.4	18.5	12.3	5.6	-0.6
		MCDB	-1.5	0.7	6.8	16.4	20.9	24.7	27.1	26.3	23.0	16.4	7.8	0.3
	10%	WB	-4.2	-2.3	2.6	9.0	14.6	18.7	21.5	20.2	17.0	10.2	3.4	-2.2
		MCDB	-3.6	-1.6	3.9	14.3	19.7	23.2	25.8	24.7	21.5	13.8	5.3	-1.5
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	9.6	9.3	11.2	12.3	11.0	11.7	12.7	12.2	9.8	8.4	8.2
		MCDBR	9.4	8.2	11.1	16.3	16.4	13.9	13.3	14.7	15.8	15.0	12.1	9.5
	5% WB	MCWBR	8.7	7.3	8.2	9.5	8.0	7.1	7.3	7.6	8.5	9.1	8.9	8.4
		MCDWR	9.3	8.0	10.7	14.6	12.8	11.4	11.9	12.7	13.1	13.8	11.3	9.1
(40) Clear-Sky Solar Irradiance	taub	taub	0.273	0.300	0.305	0.353	0.370	0.383	0.386	0.381	0.347	0.305	0.275	0.257
		taud	2.272	2.204	2.386	2.351	2.366	2.354	2.355	2.366	2.457	2.542	2.492	2.386
	Ebn at Noon	Ebn at Noon	784	841	912	895	890	877	868	856	855	846	798	769
		Edh at Noon	72	98	98	113	117	119	117	111	92	71	58	56
(44) All-Sky Solar Radiation	RadAvg	1.47	2.71	3.93	4.92	5.48	5.88	6.27	5.16	3.77	2.28	1.29	1.02	
	RadStd	0.16	0.19	0.36	0.42	0.40	0.45	0.26	0.33	0.27	0.32	0.17	0.12	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (32 neighbors)		(m)								
		N/A	N/A	N/A	N/A	-0.87	-1.03	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page